



400 Gb/s O-band silicon photonic transmitter for intra-datacenter optical interconnects

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Abstract: We present and experimentally demonstrate a silicon photonic (SiP)-based four-lane 400 Gb/s transmitter for fiber-rich intra-datacenter optical interconnects. Four parallel SiP series push-pull traveling wave Mach-Zehnder modulators (MZMs) operating in the O-band are used in the transmitter. The MZMs have an average electro-optic (EO) bandwidth of approximately 30 GHz at 3 V reverse bias voltage. To assess the parallel operation, we measure the EO crosstalk between the four MZMs, where the EO crosstalk between the closest MZMs is below -17 dB over 50 GHz bandwidth. Then, we use a four-channel digital-to-analog converter (DAC) to simultaneously drive the MZMs and characterize the performance of the transmitter versus various parameters. Results reveal that 53 Gbaud pulse amplitude modulation over 4-levels (PAM4), i.e., 100 Gb/s net rate, per lane can be received at a bit error rate (BER) below the KP4-forward error correction (KP4-FEC) threshold of 2.4×10^{-4} using only a 5-tap feed-forward equalizer (FFE) at the receiver. In addition, we show that 53 Gbaud and 64 Gbaud PAM4 per lane can be received at a BER below the KP4-FEC and 7% hard decision FEC (HD-FEC), respectively, using a driving voltage swing below 1.8 Vpp. To the best of our knowledge, these are the best results for 100 Gb/s PAM4 using a single electrode SiP TWMZM with a lateral PN junction in a multi-project wafer process. Finally, we show that the BER is still below the KP4-FEC at maximum crosstalk for all lanes, and an aggregate rate of 400 Gb/s can be achieved at an average BER of approximately 1×10^{-4} .

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1. Introduction

Since 2008, peer-to-peer traffic ceased to dominate the Internet traffic and most Internet traffic has originated or terminated in a datacenter [1]. The global datacenter IP traffic will continue to grow in the foreseeable future owing to the unprecedented increases in application driven traffic demand such as video streaming, online gaming, and cloud-based storage and services. The datacenter traffic is forecasted to grow three fold in the next five years to reach 20.6 zettabytes by 2021, where more than 70% of this data traffic stays within the datacenter [1]. To cope with such increases, significant research and development efforts have been directed towards intra-datacenter optical interconnects operating over single mode fiber (SMF) links ranging from 500 m to few kilometers. Currently, 100 Gb/s transceivers based on 4 lanes $\times$ 25 Gb/s non-return to zero (NRZ) are being shipped in volume and deployed in different service providers. The four lanes are 4 parallel single mode (PSM) fibers for the 500 m reach and 4 wavelength-division multiplexing (WDM) channels for the 2 and 10 km reaches on the LAN-WDM grid, i.e, 800 GHz spacing, or coarse WDM (CWDM) grid, i.e, 20 nm spacing [2, 3]. Also, the next generation of Ethernet optical transceivers will operate at 200 Gb/s and 400 Gb/s [4]. Scaling from 100 Gb/s to 200 or 400 Gb/s requires the increase of single lane bitrate and/or number of lanes.

Increasing the single lane bitrate requires the increase of the symbol rate and/or the modulation format order. Several ≥ 100 Gb/s single carrier results have been recently reported using pulse amplitude modulation (PAM), dual polarization PAM, and discrete multi-tone (DMT) [5–9]. In 2017, the IEEE standardized the 400GBase Ethernet specifications where 4-level PAM (PAM4) has been selected as the modulation format for the SMF optical links [10]. The standard over 500 m reach of SMF is: 50 Gbaud PAM4 $\times$ 4 PSM fibers. On the other hand, eight WDM lanes each operating at 25 Gbaud with 800 GHz spacing is used for the 2 and 10 km reaches [10].

The silicon photonics (SiP) platform, among different platforms, is used to build optical transceivers targeting different reaches such as intra- and inter- datacenter ranging from 500 m to 20 km, metro-links, and even long-haul communications. This is attributed to the potential to build high volume, compact, high yield, high performance, and low cost complementary metal oxide semiconductor (CMOS) compatible devices. In the last decade, a plethora of SiP designs has been demonstrated including passive, active, and more complex photonic integrated circuits [11–18].

Recently, few ≥ 400 Gb/s demonstrations have been reported on different lanes using PSM or WDM [19]. In [20], the first real-time transmission of 400 Gb/s ($8\lambda \times 50$ Gb/s) PAM-4 signals for datacenter interconnects up to 100 km of SMF is successfully demonstrated using discrete components and bulk modulation. Also, interoperability using 400 Gb/s CFP8 modules have been presented in [21], where 8 WDM channels on the LAN-WDM grid are modulated with 25 baud PAM4. In [22], 465 Gb/s net rate has been achieved using four commercial 25 Gb/s external modulated lasers (EMLs) on the LAN-WDM grid. However, equalization, post filtering, and maximum likelihood sequence estimation are needed to compensate for the induced inter-symbol interference (ISI) due to the EML limited bandwidth used in [22]. Moreover, 400 Gb/s has been achieved using a 4λ CWDM transmitter optical sub-assembly (TOSA) based on EML technology [23]. The SiP platform has been used in [24] where a 400 Gb/s transmitter using four traveling wave Mach-Zehnder modulators (TWMZMs) is presented. However, the demonstration is based on the DMT modulation format and relatively complex digital signal processing is needed at both the transmitter and the receiver. Moreover, each MZM is tested individually and crosstalk between MZMs was not reported which is expected to degrade the performance of the transmitter. Furthermore, an eight-lane hybrid multi-chip module comprising InP lasers, SiP MZMs, and parallel SMFs, all connected via photonic wire bonds and achieving 400 Gb/s aggregate bitrate has been presented [25]. However, the demonstration is in the C-band and each lane is tested individually using RF probes while crosstalk between the modulators is expected to degrade the performance when all lanes are operated simultaneously [26].

In this paper, we present an O-band four-lane 400 Gb/s transmitter using four parallel SiP TWMZMs. The transmitter is designed for intra-datacenter optical interconnects where it is fiber rich and PSM fibers are used. The measured average electro-optic (EO) bandwidth for the MZMs is approximately 30 GHz at 3 V DC reverse bias voltage. To enable parallel operation, minimal crosstalk should exist between the modulators. The measured EO small-signal crosstalk between the closest MZMs is below -17 dB over 50 GHz bandwidth. The crosstalk decreases below -30 dB for 750 μm spacing and more. Then, we report the performance of the transmitter in a transmission test-bed versus number of receiver equalizer taps, driving voltage swing, crosstalk voltage swing, bitrate, reach, received signal power, and modulation format. Results show that 53 Gbaud PAM4 per lane can be received at a bit error rate (BER) below the KP4 forward error correction (KP4-FEC) threshold of 2.4×10^{-4} using only a 5-tap feed forward equalizer (FFE) at the receiver when the RF signals on the other lanes are disabled. Moreover, we show that 53 and 64 Gbaud PAM4 per lane can be received at a BER below the KP4-FEC and the 7% hard decision-FEC (HD-FEC) (3.8×10^{-3}) thresholds, respectively, using only 1.8 V_{pp} drive voltage swing and 11 tap-FFE. To the best of our knowledge, this is the best result for 100 Gb/s net rate per SiP TWMZM with a lateral PN junction in a multi-project wafer (MPW) run using only 1.8

Vpp and 11-tap FFE. In addition, we show the effect of crosstalk from the other channels on the tested channel, and conclude that a slight degradation occurs in the BER when the crosstalk voltage swing is above 2 Vpp. However, the BER is still below the KP4-FEC at maximum crosstalk for all lanes, where 400 Gb/s aggregate net rate can be transmitted at an average BER of approximately 1×10^{-4} . Finally, we compare the performance of PAM2, PAM4, and PAM8 modulation formats, where we show that 100 Gb/s per lane can be achieved using the same transmitter running at 35 Gbaud PAM8.

The rest of this paper is organized as follows. In section 2, the device details are explained. The small-signal characterization of the device is presented in section 3. In section 4, the experimental setup for the large-signal modulation is introduced and the experimental results are presented. Finally, the paper is concluded in section 5.

2. Design and fabrication

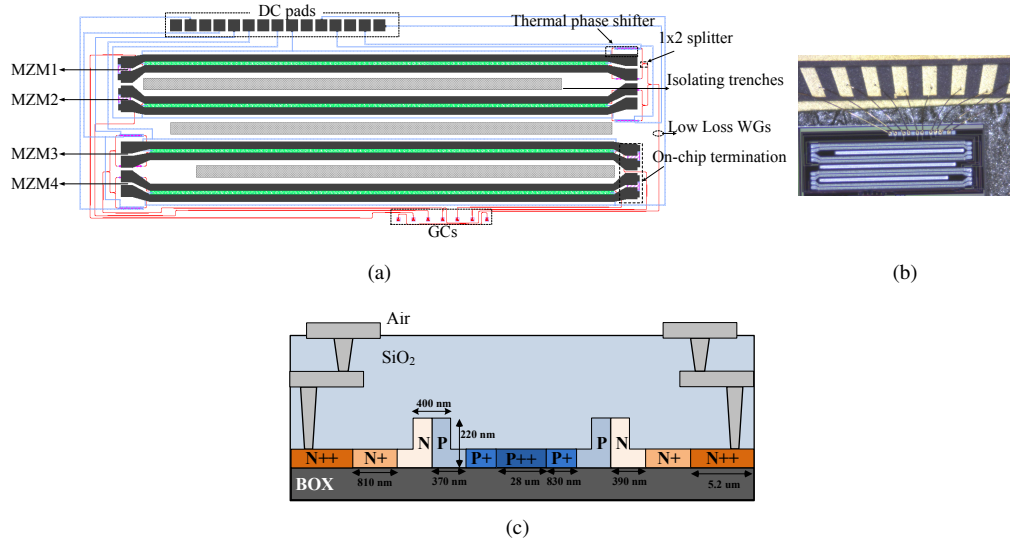


Fig. 1: (a) Layout schematic for the SiP transmitter, (b) image of the die wirebonded to a chip carrier, and (c) TWMZM cross-section.

The 400 Gb/s transmitter was fabricated in a MPW run at the advanced micro foundry (AMF) on a silicon-on-insulator (SOI) wafer with a 220-nm-thick top silicon layer, a 2- μ m-thick buried oxide (BOX) layer, and a high-resistivity 750 Ω -cm silicon substrate using 193 nm lithography. The TWMZM cross-section is shown in Fig. 1(c), where the electrode design is similar to our previous TWMZM designs in [27, 28]. Small modifications were made on the p-n junction geometry to enable O-band operation such as decreasing the waveguide width to 400 nm. All the modulators are 4.35 mm in length from pad center to pad center with a phase shifter length of $\sim$ 4 mm. Owing to the series push-pull (SPP) configuration, the microwave losses decrease and the modulation bandwidth is improved compared to the conventional dual differential drive scheme. In addition, the transmitter's driver circuit is simplified due to the need of one driving RF signal per modulator.

Figure 1(a) shows the layout schematic of the SiP transmitter. The continuous wave (CW) laser is coupled to the SiP chip using a grating coupler at the input, where it is then split by three low loss Y-branches to feed the four MZMs [29]. All the MZMs are identical and the travelling wave electrodes are terminated using on-chip 50 ohm terminations. The MZMs have approximately

360 μm spacing center-to-center. Also, 3 μm wide waveguides are used for routing to decrease the routing losses due to scattering from the waveguide sidewall roughness. The MZMs are balanced and a thermal phase shifter is added to one of the arms of each modulator to bias the modulators at the quadrature point. Moreover, isolating deep trenches are added between the MZMs to decrease the crosstalk between the modulators. The outputs of the four MZMs are connected to four grating couplers to enable parallel operation. The DC connections for the transmitter were wirebonded to a chip carrier as shown in Fig. 1(b).

3. Device characterization

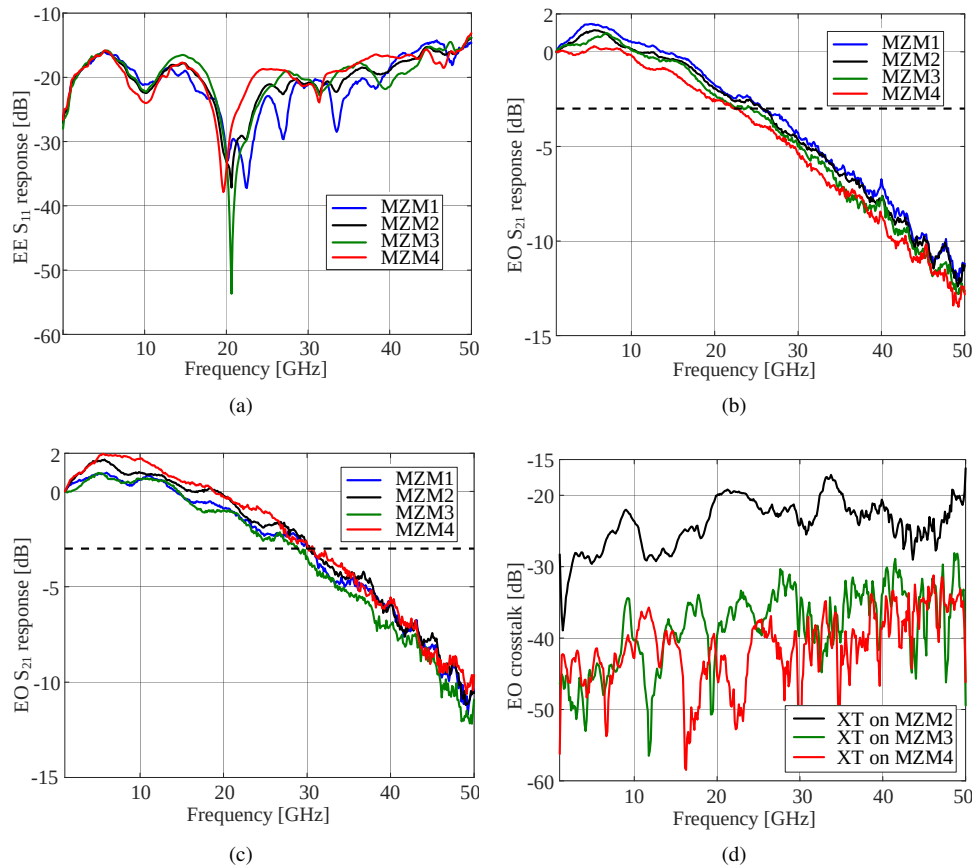


Fig. 2: (a) EE S_{11} response for the four MZMs, (b) and (c) EO S_{21} response for the MZMs at 0 V and 3 V DC bias, respectively, and (d) EO crosstalk between MZM1 (aggressor) and MZM2-4 (victims).

In this section, we present the insertion loss (IL) and small-signal characterization results for the 400 Gb/s transmitter. The fiber-to-fiber IL measured at maximum transmission was approximately 21.2 dB from the input to the output of one of the modulators. The ILs breakdown is as follows: ~ 9 dB from the grating coupler pair, 6.6 dB from the splitters (splitting and excess losses), 4.1 dB modulator IL, and 1.5 dB routing losses. Using low loss edge couplers and more optimized routing will decrease the ILs by 6 - 8 dB [30–32].

A 50 GHz Keysight lightwave component analyzer (LCA) and 50 GHz GSSG probes were used to perform the small-signal characterization for the four MZMs. Figure 2(a) shows the

measured electrical-electrical (EE) S_{11} responses for the four MZMs. The S_{11} magnitude is well below -15 dB over 50 GHz. The EO S_{21} magnitude responses for DC bias voltages of 0 V and 3 V are shown in Figs. 2(b) and 2(c) where the four MZMs have very close results as expected. The 3-dB bandwidth is approximately 25 GHz at 0 V and increases to 30 GHz at 3 V reverse bias voltage.

Next, we characterize the crosstalk between the four MZMs in Fig. 2(d). The crosstalk between the transmission lines can be attributed to electric and magnetic radiative crosstalk and conductive substrate crosstalk. We added deep trenches between neighbor MZMs to reduce through-substrate conductive crosstalk, however, it helps to a small degree since most of the cross talk is radiative between the electrode metallic lines. This conclusion is reached by comparing our result to the results in [26], where no trenches are added and nearly same separation of more than $600\text{ }\mu\text{m}$ is needed to have negligible crosstalk penalty. Hence, we conclude that the crosstalk is mainly due to radiation between neighbor MZMs. We find the EO crosstalk between MZM1 (aggressor), and the other MZMs (victims) using GSSG probes. On the other hand, SGGS probes, which were not available at the time of the experiment, are expected to have less crosstalk compared to the GSSG probes as shown in [33]. To measure the EO crosstalk, the RF signal from the LCA is launched into the input of MZM1 and the optical output of the victim MZM is connected to the optical port of the LCA. Then, the measured crosstalk is normalized to the victim's EO response and plotted in Fig. 2(d). It can be observed that the crosstalk between MZM1 and MZM2 is below -17 dB over the entire 50 GHz range, and increases with frequency. Hence, we expect that the crosstalk will have more impact on the performance at higher bitrates. Although the crosstalk value here is relatively low, we show in the next section that it is not negligible and the MZM spacing should be increased more than the used $360\text{ }\mu\text{m}$ to decrease the penalty when the MZMs are simultaneously modulated. In addition, the crosstalk between MZM1 and both MZM3 and MZM4 which are spaced by $\sim 750\text{ }\mu\text{m}$ and $1100\text{ }\mu\text{m}$, respectively, is well below -30 dB and can be neglected. This agrees with the results in [26] where negligible penalty is achieved at $600\text{ }\mu\text{m}$ spacing between the SiP modulators.

4. Large-signal experimental setup and results

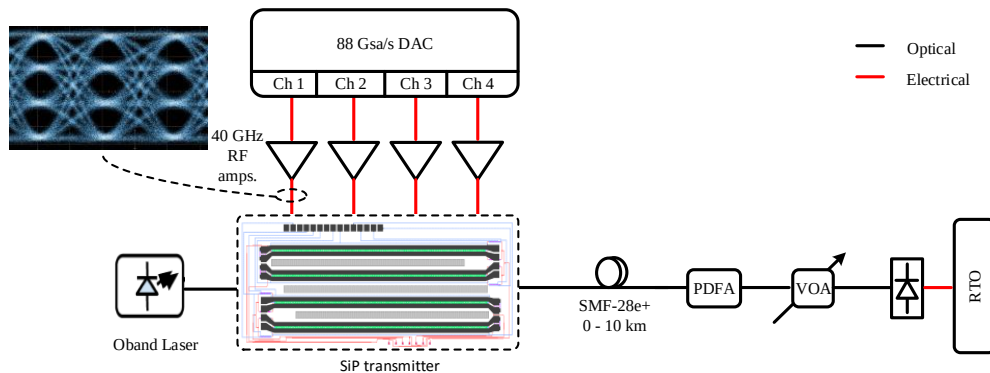


Fig. 3: Experimental setup used for the 400G PSM transmitter testing. Inset: 53 Gbaud PAM4 RF signal out of the amplifier. DAC: digital-to-analog converter, PDFA: praseodymium-doped fiber amplifier, SMF: single mode fiber, VOA: variable optical attenuator, and RTO: real time oscilloscope.

Figure 3 shows the experimental setup used to characterize the 400 Gb/s transmitter. An O-band laser launches a 16 dBm CW light at 1310 nm wavelength to the SiP chip using the input grating coupler. An 8-bit digital-to-analog converter (DAC) running at 88 GSa/s is used to

generate four PAM4 signals. Then, the output RF signals from the four channels of the DAC are amplified using four 40 GHz RF amplifiers. A 53 Gbaud PAM4 eye diagram after the RF amplifier for one of the driving lanes is shown in the top-left inset of Fig. 3(a). At the transmitter side, we only pre-compensate the response of the DAC and RF amplifier. No pre-emphasis or non-linearity pre-compensation is used for the MZMs, and only simple level shifting for the inner levels of the PAM4 signal is done at some bitrate values. The driving signals are applied to the four modulators using 50 GHz GSSG probes. The modulated optical signal is then launched into various lengths of SMF (Corning SMF-28e+) covering reaches ranging from 500 m to 10 km. Also, a praseodymium-doped fiber amplifier (PDFA) is used to provide sufficient signal power to the 50 GHz photodetector which has no transimpedance amplifier stage. To sweep the received signal power, a variable optical attenuator (VOA) is added before the receiver. Finally, the signal out of the photodetector is sampled at 160 GSa/s by a 62 GHz real time oscilloscope (RTO) and stored for offline processing. The offline processing includes: resampling, FFE, symbol de-mapping, and bit error counting.

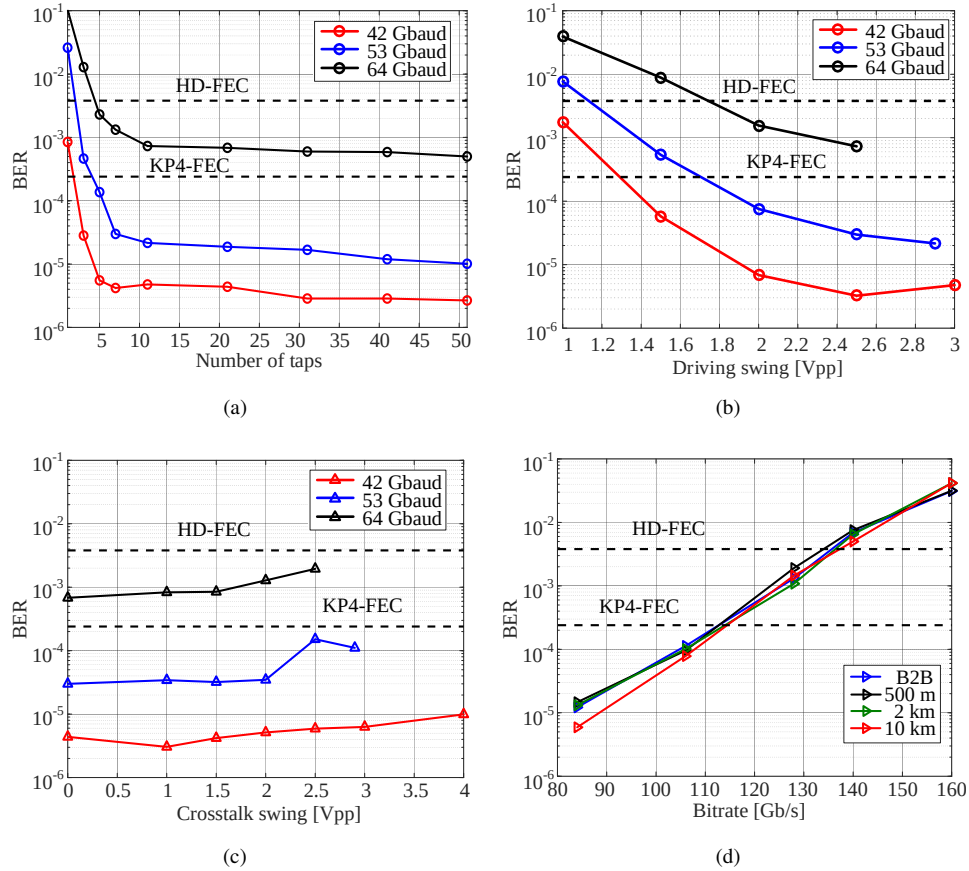


Fig. 4: (a) BER performance versus number of receiver FFE taps for different symbol rates, (b) BER performance versus driving voltage swing without crosstalk, (c) crosstalk impact on BER performance at different symbol rates, and (d) BER performance versus bitrate for a single lane in presence of crosstalk over different reach values at constant received signal power.

First, we measure the RF V_{π} using a 10 Gb/s on-off keying (OOK) driving signal. We launch the RF signal to the MZM, and monitor the optical eye diagram on the digital sampling scope.

By sweeping the driving voltage swing, we find the RF V_{π} to be approximately 6 V. Hence, the measured BW/V_{π} is approximately 5 GHz/V. Next, we show the BER performance results for one of the MZMs and similar results were found for the other MZMs versus number of receiver equalizer taps, driving voltage swing, crosstalk swing voltage, bitrate, and reach. In Fig. 4(a), the BER performance versus the number of receiver equalizer taps is reported at different symbol rates of PAM4 modulation format in the absence of crosstalk from the other MZMs. To reiterate here, only the response of the DAC and RF amplifiers are pre-compensated at the transmitter and the rest of the chain including the modulator and PD is left at the receiver side where a FFE is used and the taps are found adaptively using the least mean squares algorithm (LMS). For the 53 Gbaud PAM4, only a 5-tap FFE at the receiver is needed to achieve a BER below the KP4-FEC threshold of 2.4×10^{-4} . Increasing the number of taps further to 41 taps improves the BER performance to reach $\sim 1 \times 10^{-5}$ at the expense of increasing the complexity. The optimal number of taps is found to be approximately 11 taps, where further increases has a small improvement in the BER. Also, we can observe a similar trend for the 42 Gbaud and 64 Gbaud symbol rates. However, we observed that a BER below the HD-FEC can only be achieved for the 64 Gbaud curve, this can be attributed to the limited driving swing, as will be discussed next, as well as the increased inter-symbol interference (ISI). For the rest of the results, we fix the number of taps at 11 taps.

In Fig. 4(b), we study the BER performance versus the driving voltage swing at different symbol rates for the PAM4 modulation format in the absence of crosstalk from the other MZMs. Less than 1.8 Vpp is needed to achieve a BER below the KP4-FEC threshold at 100 Gb/s net rate. To the best of our knowledge, this is the first time to achieve 100 Gb/s net rate using a MZM in a MPW process driven by less than 1.8 Vpp driving swing and few taps at the receiver side. Increasing the driving swing further improves the BER performance to reach approximately 2×10^{-5} at 2.9 Vpp voltage swing. Similarly, less than 1.8 Vpp is needed to achieve a BER below the HD-FEC for the 64 Gbaud case. The maximum voltage swing at 64 Gbaud is 2.5 Vpp which is the maximum achievable voltage from the RF amplifier at such symbol rate. Increasing the driving swing above 3 Vpp together with more FFE taps, we expect to reach a BER below the KP4-FEC for the 64 Gbaud symbol rate. For the rest of the results, the driving voltage swing of the lane under test is fixed at the maximum achievable voltage.

Then, we enable the RF signal on all MZMs, and study the effect of the crosstalk voltage swing of the three lanes on the lane under test in Fig. 4(c). At 53 Gbaud, it can be observed that driving all lanes simultaneously has a negligible effect on the BER up to a crosstalk voltage swing of 2 Vpp. The BER performance deteriorates when the crosstalk driving swing is increased to 2.5 Vpp. However, the BER is below the KP4-FEC at all crosstalk values. From that we conclude that the current 350 μ m modulators spacing is not sufficient to completely mitigate the effects of crosstalk. Figure 4(d) presents the BER performance versus bitrate over different reach values at 7 dBm received signal power for a single MZM while all MZMs are simultaneously modulated. In this figure, the number of taps is fixed at 11 taps to decrease the system complexity and the swing is the maximum available swing after the RF amplifier. For example, at 53 Gbaud the voltage swing is 2.9 Vpp, and reaches below 1.2 Vpp for the 80 Gbaud signal. As expected, the system is loss-limited, and hence the BER is nearly constant with reaches up to 10 km of SMF at equal received signal power. The BER for 100 Gb/s signal is approximately 1×10^{-4} after 10 km propagation. The BER degrades with further increases in the bitrate to reach 4×10^{-3} at 140 Gb/s due to both the ISI and the limited voltage swing. Hence, increasing the voltage swing and using a stronger equalizer, we expect to operate below the FEC threshold.

Next, we characterize the performance of the four lanes of the 400 Gb/s transmitter. Figure 5 presents eye diagrams for the four MZMs of the transmitter driven by 53 Gbaud PAM4 signal after receiver DSP. The eye diagrams show clear open eyes from all lanes at the receiver side at 100 Gb/s net rate per lane while all lanes are simultaneously modulated. The BER performance

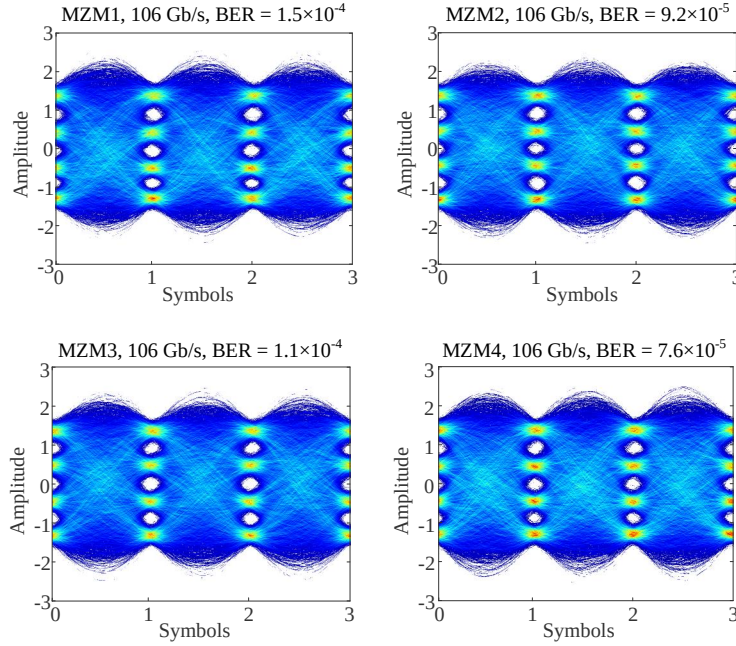


Fig. 5: Eye diagrams for the four MZMs simultaneously modulated obtained after receiver DSP at 100 Gb/s net rate.

for all MZMs versus bitrate in the B2B case at 7 dBm received signal power is shown in Fig. 6(a). We also add the BER performance using a reference Lithium Niobate modulator to Fig. 6(a). The reference modulator has a bandwidth of 30 GHz and V_π of 5 V. It can be observed that all MZMs have a close performance as expected. Also, we can achieve 106 Gb/s per lane, i.e., 400 Gb/s aggregate net rate, below the KP4-FEC threshold using 2.9 Vpp drive voltage and 11 tap-FFE. To the best of our knowledge, this presents the first demonstration of 400 Gb/s in the O-band using simultaneously modulated SiP based MZMs driven by below 3 Vpp and simple DSP targeting intra-datacenter optical interconnects. The estimated power consumption to achieve 100 Gb/s net rate per lane at a BER below the KP4-FEC threshold is 9 mw and 23.36 mw with and without crosstalk from other lanes, respectively, excluding the power consumption from the thermal phase shifters for the biasing. The energy per bit for the four lanes for an aggregate rate of 400 Gb/s is 0.9344 pJ/bit. Moreover, if more spacing between the MZMs is added, the MZMs can be driven with lower voltage and crosstalk penalty can be neglected yielding energy per bit for the four lanes of only 0.36 pJ/bit.

Figure 6(b) shows the received signal power of all lanes with and without crosstalk running at 53 Gbaud PAM4. It can be observed that approximately an average of 3 dBm and 4 dBm received signal power is needed to achieve a BER below the KP4-FEC threshold with and without crosstalk, respectively. Hence, ~1 dB penalty occurs due to simultaneous modulation at high received signal powers, whereas the penalty is negligible at lower received power values since the performance is dominated by other noise sources such as receiver noise.

Finally, we compare the BER performance versus bitrate for PAM2, PAM4, and PAM8 modulation formats for one lane while other lanes are enabled as shown in Fig. 7(a). To reiterate here, the modulator transfer function is not pre-compensated at the transmitter side. The number of receiver FFE taps is fixed at 11, 11, and 31 taps for PAM2, PAM4, and PAM8, respectively. Eye diagrams after receiver DSP for 53, 106, and 105 Gb/s signals using PAM2, PAM4 and

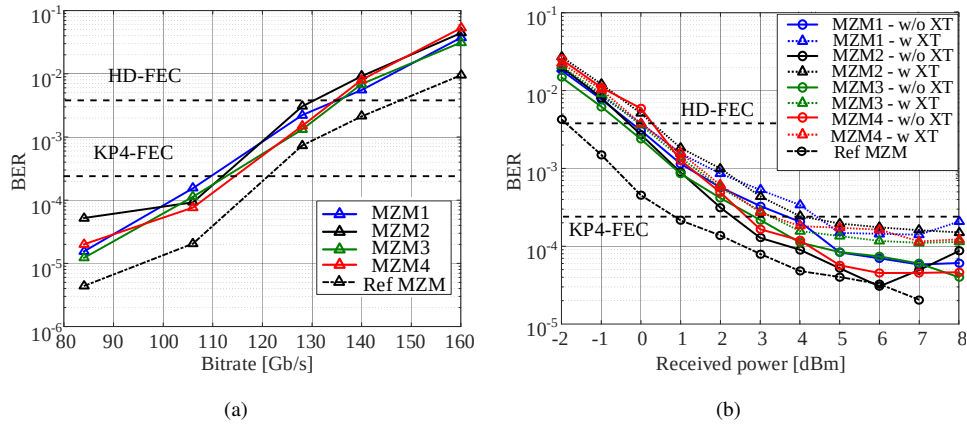


Fig. 6: BER versus bitrate for the four MZMs simultaneously modulated in the B2B case, and (b) BER versus received signal power with and without presence of crosstalk from other lanes. Results from a reference lithium niobate MZM are added for comparison.

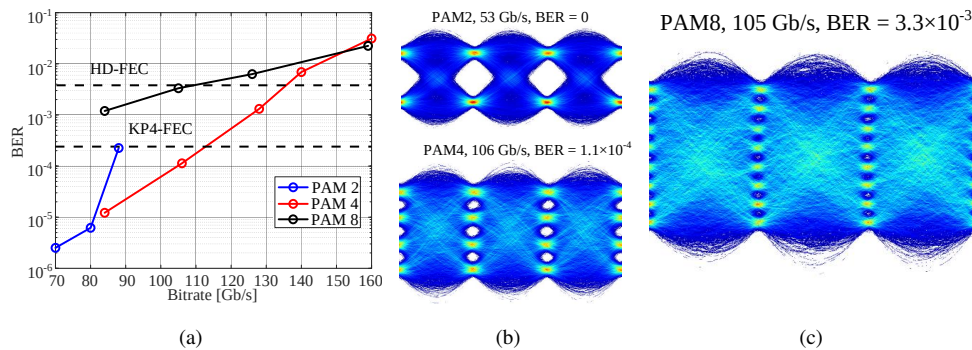


Fig. 7: (a) BER versus bitrate for different modulation formats in the B2B case, and (b-c) eye diagrams after receiver DSP for PAM2, PAM4 and PAM8 modulation formats running at 53, 53, and 35 Gbaud, respectively.

PAM8 modulation formats, respectively, are shown in Figs. 7(b) and 7(c). For the PAM2 signal, a bitrate up to 88 Gb/s can be received at a BER below the KP4-FEC threshold. It is interesting to notice that to achieve 100 Gb/s per lane while still having low complexity, we need to switch to PAM4 modulation format instead of PAM2. However, a higher BER is achieved and a relatively strong FEC is needed to switch from PAM2 to PAM4 modulation format. Finally, we expect for higher target bitrates another transition to PAM8 modulation format that operate at lower symbol rate at the expense of utilizing a stronger FEC and equalization.

5. Conclusion

We present the design and experimental demonstration of a SiP MZM based four lane 400 Gb/s for intra-datacenter optical interconnects. We report the device details, small-signal, and large-signal characterization of the transmitter. The measured EO bandwidth and RF V_{π} of the MZMs are approximately 30 GHz and 6 V, respectively. Also, we show the EO crosstalk between the four MZMs, and conclude that more than 750 μm spacing is needed for parallel operation. For large-signal modulation, we characterize the performance of the transmitter versus several

parameters. Using only a 5-tap receiver FFE, 53 Gbaud PAM4 per lane can be received at a BER below the KP4-FEC threshold without crosstalk from the other MZMs. Moreover, we show that the MZMs can be driven with a driving voltage swing below 1.8 Vpp and still achieve a BER below the KP4-FEC threshold for a 53 Gbaud PAM4 signal. Although, several 100 Gb/s demonstrations have been published to date based on SiP MZMs, we believe this is the best result for a MZM with a lateral PN junction in terms of driving voltage swing and equalization complexity to the best of our knowledge. Moreover, we demonstrate the first demonstration of a simultaneous modulation of a 4-lane transmitter running at an aggregate rate of 400 Gb/s with an average BER of approximately 1×10^{-4} for all lanes. Less than 5 dBm received signal power is needed to achieve a BER below the KP4-FEC for the four MZMs driven by 53 Gbaud PAM4 each with a 1 dB penalty compared to without crosstalk case. Finally, we compare the performance of PAM2, PAM4, and PAM8 modulation formats, where we show that 100 Gb/s per lane can also be realized using the same transmitter running at 35 Gbaud PAM8.

Acknowledgments

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